

Recall from last class:

- **Theorem:** Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases of a vector space V . Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.
- We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . In general

$$P = [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}].$$

Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

- Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \dots, c_m\}$. Given a linear transformation $T: V \rightarrow W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix M such that $[T(x)]_{\mathcal{C}} = M[x]_{\mathcal{B}}$ for all $x \in V$. We call M the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$** . In general $M = [[T(b_1)]_{\mathcal{C}} \ \dots \ [T(b_n)]_{\mathcal{C}}]$.
- **Diagonal Representation Theorem:** Suppose $A = PDP^{-1}$ where D is a diagonal matrix and $T(x) = Ax$. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed by the columns of P , then D is the matrix of T relative to $\mathcal{B}$.
- If A and B are $n \times n$ matrices, then A is **similar** to B if there exists an invertible matrix P such that $A = PBP^{-1}$.

Definition 1. If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

1. Let $A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Determine if v is an eigenvector of A and find the corresponding eigenvalue.

Definition 2. If λ is an eigenvalue of an $n \times n$ matrix A , then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ .

2. Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$, then $\lambda = 2$ is an eigenvalue. Find the corresponding eigenspace.

Theorem 1: The eigenvalues of a triangular matrix are the diagonal entries.

3. Prove the theorem.

Theorem 2: If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then $v_1, \dots, v_p$ are linearly independent.

4. Prove the theorem

5. For a general $n \times n$ matrix A , how can we find the eigenvalues?

6. Let $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$ Find the eigenvalues of A and corresponding eigenspaces.

7. Prove that λ is an eigenvalue of A if and only if λ is an eigenvalue for A^T .

Definition 3. If A is an $n \times n$ matrix, then the characteristic polynomial of A , denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

Note: λ is an eigenvalue of A if and only if λ is a zero of the characteristic polynomial.

8. Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 6 \\ 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ Find the characteristic polynomial of A .

9. If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).

10. True or False.

- (a) If two matrices have the same eigenvalues, then they are similar.
- (b) If two matrices are row equivalent, then they have the same eigenvalues.
- (c) If two matrices are row equivalent, then they are similar.

Eigenvalues and Eigenvectors – Answers / Solutions

1. Let $A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, then

$$Av = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = (-2)v.$$

Thus multiplying v by A is just scaling v by -2 , so v is an eigenvector corresponding to the eigenvalue of -2 .

2. Recall the eigenspace corresponding to the eigenvalue of $\lambda = 2$ is given by $E_2 = N(A - 2I_3)$. Note

$$A - 2I_3 = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -16 & \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$N(A - 2I_3) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

3. *Proof.* Let A be an upper-triangular matrix, then we want to know what $\lambda \in \mathbb{R}$ allow for non-trivial solutions to $(A - \lambda I_n)x = 0$. Thus we have

$$A - \lambda I_n = \begin{bmatrix} a_{11} - \lambda & & * \\ & \ddots & \\ 0 & \cdots & a_{nn} - \lambda \end{bmatrix}.$$

Thus $(A - \lambda I_n)x = 0$ has non-trivial solutions if and only if $A - \lambda I_n$ is not invertible by the Invertible Matrix Theorem. Again by the Invertible Matrix Theorem, $A - \lambda I_n$ is not invertible is equivalent to $\det(A - \lambda I_n) = 0$. We proved the determinant of an upper-triangular matrix is the product of the diagonal entries. Thus we have

$$0 = \det(A - \lambda I_n) = (a_{11} - \lambda) \cdots (a_{nn} - \lambda).$$

Thus λ is an eigenvalue for A if and only if λ is one of $a_{11}, \dots, a_{nn}$. □

4. *Proof.* We prove the result by contradiction. Suppose $v_1, \dots, v_p$ are linearly dependent. Since $v_1 \neq 0$, there is a smallest $r > 1$ such that

$$v_r = c_1 v_1 + \cdots + c_{r-1} v_{r-1}.$$

Applying A to both sides and simplifying, we get the following

$$\begin{aligned} Av_r &= A(c_1 v_1 + \cdots + c_{r-1} v_{r-1}) \\ Av_r &= c_1 A v_1 + \cdots + c_{r-1} A v_{r-1} \\ \lambda_r v_r &= c_1 \lambda_1 v_1 + \cdots + c_{r-1} \lambda_{r-1} v_{r-1} \\ \lambda_r (c_1 v_1 + \cdots + c_{r-1} v_{r-1}) &= c_1 \lambda_1 v_1 + \cdots + c_{r-1} \lambda_{r-1} v_{r-1}. \end{aligned}$$

Rearranging the equation, we get

$$0 = (c_1\lambda_1 - \lambda_r c_1)v_1 + \cdots + (c_{r-1}\lambda_{r-1} - \lambda_r c_{r-1})v_{r-1}.$$

Since $v_1, \dots, v_{r-1}$ are linearly independent, we get

$$(c_1\lambda_1 - \lambda_r c_1) = \cdots = (c_{r-1}\lambda_{r-1} - \lambda_r c_{r-1}) = 0.$$

Since $\lambda_1, \dots, \lambda_r$ are distinct, we must have $c_1 = \cdots = c_{r-1} = 0$. This implies $v_r = 0$, but eigenvectors must be non-zero. Thus we have a contradiction. $\square$

5. We've seen several times now that λ is an eigenvalue for A if and only if $A - \lambda I_n$ is not invertible. Thus we can find the eigenvalues of A by finding all values of λ such that $\det(A - \lambda I_n) = 0$.

6. Let $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$, then

$$p_A(\lambda) = \det \begin{bmatrix} 3 - \lambda & 2 \\ 3 & 8 - \lambda \end{bmatrix} = (3 - \lambda)(8 - \lambda) - 6 = (\lambda - 9)(\lambda - 2).$$

Thus the eigenvalues of A are 2 and 9.

$\lambda = 2$ Here we have $A - 2I_2 = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$. Thus

$$E_2 = N(A - 2I_2) = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}.$$

$\lambda = 9$ Here we have $A - 9I_2 = \begin{bmatrix} -6 & 2 \\ 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix}$. Thus

$$E_9 = N(A - 9I_2) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

Thus

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 9 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}^{-1}.$$

7. *Proof.* Since $p_{A^T} = \det(A^T - \lambda I_n) = \det(A - \lambda I_n)^T = \det(A - \lambda I_n) = p_A$, we get A and A^T have the same characteristic polynomials. Thus they have the same eigenvalues with the same algebraic multiplicities. $\square$

8. From the previous theorem, the characteristic polynomial is given by

$$p_A(\lambda) = (\lambda - 1)(\lambda - 5)^2(\lambda - 2).$$

Thus 1 and 2 are eigenvalues with algebraic multiplicities 1 and 5 is an eigenvalue with algebraic multiplicity 2.

9. *Proof.* Since A and B are similar, there exists an invertible matrix P such that $A = PBP^{-1}$. Then

$$p_A(\lambda) = \det(A - \lambda I_n) = \det(PBP^{-1} - \lambda PP^{-1}) = \det P p_B(\lambda) \det P^{-1} = p_B(\lambda).$$

Thus A and B have the same characteristic polynomials and hence they have the same eigenvalues with the same algebraic multiplicities. $\square$

10. (a) **False.** Let $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and let $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then both A and B are upper triangular, and hence the eigenvalues are given by the entries on the main diagonal. Thus both A and B have eigenvalue 0 with algebraic multiplicity 2. However, for any invertible matrix P , we get $PAP^{-1} = A$. Thus A and B cannot be similar.
- (b) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and let $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then $A \sim B$, but they have different eigenvalues.
- (c) **False.** Again let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and let $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then $A \sim B$. But they have different eigenvalues, so by the contrapositive of problem 9, we get A and B cannot be similar.